Track Stack: Technical Architecture & System Specification

Comprehensive guide on architecture, technology stack, security cryptography, and implementation details.

1. Architecture Overview & Core Stack

Track Stack is engineered as a high-integrity, production-grade Club Activity Management Platform for Mirai School of Technology (MSOT). The platform unifies club governance, attendance check-ins, deliverable versioning, rubric evaluations, events, duty leaves, and verifiable transcripts into a reactive App Router architecture.

Layer / Technology	Primary Purpose	Why We Chose It
Next.js 16 (App Router)	Full-Stack Web Framework	Server components, fast API route
TypeScript (Strict)	Type System	Zero-any policy, end-to-end type s
React 19	Reactive UI Runtime	Server actions, concurrent renderi
Tailwind CSS	Utility Design System	High-contrast dark navy tokens (#0
Supabase (PostgreSQL)	Relational Database	Hosted PostgreSQL with Row Level S
pdf-lib	Vector PDF Engine	Server-side vector rendering of cr
qrcode	Matrix Code Encoding	Dynamic SVG and PNG QR generation
Google Gemini SDK	Lumi AI Assistant	Gemini 2.5 Flash model for convers
Sharp	High-Fidelity Imaging	Image processing, circular avatars
Zod	Schema Validation	Strict runtime boundary validation

2. Cryptographic Security & Anti-Proxy Architecture

To solve the endemic challenge of attendance fraud, proxy check-ins, and forged paper certificates in educational institutions, Track Stack implements a five-tier cryptographic security engine:

* Rotating Time-Stepped HMAC Tokens: Every 25 seconds, the coordinator projector view derives a fresh token: HMAC-SHA256(sessionId:clubDayId:exp:salt:otp, HMAC_SECRET). Expired tokens are rejected instantly by server-side guards, permanently eliminating WhatsApp screenshot forwarding.
* Haversine GPS Geofencing: The client scanner submits device GPS coordinates. The server calculates Great-Circle distance using the Haversine formula against campus coordinates (28.5355 N, 77.3910 E), rejecting check-ins outside the 300m radius.
* Hardware Device Fingerprinting: Combines browser user-agent, viewport geometry, and hardware metrics into a deterministic 8-character hex digest. Multiple check-ins from the same device flag the record with device_reuse_flag = true for coordinator review.
* Cryptographically Signed Certificates: Certificates embed an HMAC-SHA256 digital signature of canonical student and event data. Verification recomputes this signature on /verify/[id], detecting any document tampering or revocation.

3. Page-by-Page Technical Engineering

Each route in Track Stack is built with dedicated architectural principles:

* Landing Page (/): Public hero showcasing university tagline, theme initializer script, and live Public Certificate Verification lookup form querying /verify/[id].
* Authentication (/login, /register): Dual email/password and workable Google Workspace OAuth selector, with 1-tap demo credentials switcher for rapid evaluation across all 5 roles.
* Analytics Dashboards (/dashboard): Role-scoped Recharts visualizations (Admin, Campus Manager, Faculty Advisor, Student Coordinator, Student) rendering attendance funnels in sub-50ms.
* Coordinator Run Deck (/activities/[id]/run): Mobile-first, one-handed operational controller with SVG rotating QR countdown ring, 6-digit OTP tabular display, and 1-tap attendance toggles with mandatory reason auditing.
* Submissions Engine (/activities/[id]/submissions): Multi-format file upload engine (PDF, PPTX, ZIP, links) with full version history archiving (v1, v2) and automated late-submission tagging.
* Rubric Evaluation (/activities/[id]/evaluate): Standardized 0-10 rubrics (Technical Depth 40%, Presentation 30%, Participation 30%) with draft score concealment until official publication.
* Clubs & Sub-Clubs Directory (/clubs, /clubs/society/[id]): Campus directory managing NexCell, Malhar, Khel Society, and Kaizen Society with its 3 specialized sub-clubs: Dev Club, CP Club, and KI.
* Events & 200+ Concurrency (/events, /events/[id]): Flagship hackathons and Olympiads with atomic team formation and waitlist promotion, stress tested for 200+ concurrent student registrations.

- Duty Leave Portal (/duty-leave): Automatic leave request dispatch on inter-campus event registration, routed to Campus Manager for approval with downloadable signed PDF letters.
- Co-Curricular Transcript (/transcript): Student-owned verified record compiling club memberships, attendance %, activity scores, and achievements with embedded verification QR code.
- Admin Audit Ledger (/admin/audit): Append-only, immutable record of attendance corrections, grade modifications, certificate revocations, and system configuration adjustments.
- Lumi AI Assistant Console (/admin/lumi & Floating Drawer): Dual-engine conversational assistant (Google Gemini SDK + scored local knowledge base) with safe read-only RLS tools and multilingual NLP.

4. Database Schema & Migration Indexing

Track Stack features 7 comprehensive PostgreSQL migrations located in supabase/migrations/:

- 001_initial_schema.sql: Core user directory, campus definitions, club registries, and basic membership tables.
- 002_phase2_schema.sql: 2-club membership constraint triggers, waitlist queues, and Club Day activity lifecycle tables.
- 003_phase3_schema.sql: Rotating check-in sessions, attendance logs, submissions with versioning, and rubric evaluations.
- 004_phase4_schema.sql: Campus events, team registrations, digital certificates, and Duty Leave approval queues.
- 005_phase5_schema.sql: Monthly reports table, audit log ledger, notifications centre, and cron automation tracking.
- 006_phase6_lumi.sql: Lumi FAQ knowledge-base schema, query analytics, and escalation support requests.
- 007_phase7_performance.sql: Compound B-Tree indexes (attendances, submissions, events, users) optimizing queries for sub-second p95 latency under 500 concurrent virtual users.

5. Development Team: Version Controllers

Track Stack was architected and developed by team Version Controllers at Mirai School of Technology:

* Ayush Chauhan (Team Leader) - GitHub: https://github.com/Ayush-git800

* Rishav Ranjan - GitHub: https://github.com/rishavgiri1106-dotcom